



Sheet 1

Project Objectives, Risks & Work Breakdown Structure (WBS)

Question 1:

Write three objectives, and one possible risk for each of the following projects inside our college:

- i- Participating in the EVER car race competition in Egypt.
- ii- Establish a society for paper recycling inside our faculty.
- iii- The student's union organize a trip to Sharm Elsheikh.

Question 2:

You are a junior student in the mechanical department of an engineering college, and you are looking for a one-month internship opportunity inside Egypt or abroad. You intend to start in the summer. Make a work breakdown structure (WBS) for the activity of “finding the best internship available”. Here are some guide tasks that you may include in your WBS. Mind that they are not listed in order.

- Prepare recommendation letters
- Choose among candidate internships
- Prepare your CV

Question 3:

Make the WBS required to sell your used car and buy a brand new one. Mention one risk for this project.

Question 4:

In our college, we need to add signs near the gates and inside the campus to guide students and visitors about the directions to different buildings and places. Make a WBS for the project. Represent the first and second level of the WBS in two different ways. Mention one risk for the process.

Sheet 2 Time Estimation

Question 1:

A project that involves several activities has effort time estimations as given in the following table. Find the estimated time duration for each activity in each of the following cases:

Activity	A	B	C	D	E
Effort time (hours)	5	3.5	7	1.5	9

- i- Percentage of regular and unplanned interruptions are 20%, and 30% respectively [A: 8.9]
- ii- Percentage of time devoted to work is 70%, and the unplanned interruptions percentage is 25% [A: 9.5]
- iii- Comment on the differences.

Question 2:

Luban workshop will upgrade some of its machines and instruments. Some of the activities of establishing the original Luban workshop are similar in nature to the new project. The following table shows the similarities and differences among these activities.

	Activity	During establishment	Previous project time (weeks)	During the upgrade
R	Painting walls	Area of 500 m ²	5	Area of 100 m ²
S	Setup of machines	15 machines	0.5	15 machines
T	Ordering for seats	24 seats	2	36 seats
V	Preparation for opening day	10 guests	1	5 guests

It is required to estimate the time of each of the 4 listed activities using the similarity and parametric estimation methods.
[S=0.5, V=1]

Question 3:

It is required to build a new educational building in the faculty of commerce of ASU. The building is designed to have 15 educational halls. A very recent project was an educational building in the faculty of law. It had 22 educational halls, and a cost of EGP 5,000,000. Assuming similarity between both projects, what is the estimated cost of the new building? [7.3M]

Question 4:

In a breakthrough project, it is required to estimate the time of an activity using the Delphi technique. Five experts were selected to give estimations in three rounds. Their estimations (in months) are recorded in the table. What is the estimated time of the activity? [31]



Question 4 (Cont.)

	Expert 1	Expert 2	Expert 3	Expert 4	Expert 5
Round 1	35	30	27	31	29
Round 2	33	30	27	33	30
Round 3	33	30	29	32	31

Question 5:

The three-point technique was used to estimate the time of some project activities. The expert estimated the values in the next table. Find the time estimate for each activity [A=43.3]

Activity	Most Likely Time (days)	Optimistic time (days)	Pessimistic Time (days)
A	42	42	50
B	5	5	5
C	25	21	25
D	25	21	35
E	4	2	6

Question 6:

The time estimate values of two activities (A & B) obtained from 5 experts in a wide-band Delphi technique are tabulated as follows.

Activity A			
Expert	Most Likely Time (hrs)	Optimistic time (hrs)	Pessimistic Time (hrs)
1	9	7	13
2	9	6	15
3	10	8	12
4	12	9	15
5	11	9	16



Question 6 (Cont.)

Activity B			
Expert	Most Likely Time (hrs)	Optimistic time (hrs)	Pessimistic Time (hrs)
1	22	20	25
2	26	25	26
3	27	25	33
4	22	19	30
5	26	23	34

- i- Find the time estimate for each activity using the wide-band Delphi technique [10.4 , 25.2]
- ii- Find the time estimate for each activity without excluding the extreme values [10.5 , 25.1]
- iii- Comment on the differences.

All final answers are given for guidance only. A student should present a complete answer including the units.



Sheet 3

Networks – Critical Path Method – Gantt Chart

Question 1:

For the following project, draw the precedence network, then find:

- i. the project completion time using the critical path method (CPM). [27 days]
- ii. Find total and free slacks of all the activities. [A:0, C:13, J:13, Q:9 ; A:0, C:0, J:10, Q:9]
- iii. Represent the project on a Gantt chart.

Activity	A	B	C	D	E	F	G	H	I
Duration (days)	2	3	1	6	5	2	2	3	2
Depends upon	----	A	A	B	D, H	D	F	B	H

(Cont.)

Activity	J	K	L	M	N	O	P	Q	R
Duration (days)	3	4	3	3	2	4	5	1	3
Depends upon	C	D, E, J	D, E, G	E, G, L	B, H	N	F, J, L	N, O	I, K, M, O, P, Q

Question 2:

A company that manufactures plastic cups and dishes for kids is planning to introduce new colors and design for their products. They predicted 20% increase for the sales; therefore, they are also planning to make an extension for the workshop area. The project manager made a list of the activities and their estimated times.

Activity	Code	Estimated Time (days)
Select new designs for production	A	2
Purchase new machines	B	30
Select the workshop extension area	C	6
Vote for new designs	D	7
Produce prototypes	E	5
Install new machines	F	7
Market research for new designs	G	10
Select new machines	H	5
Prepare extension area	K	30

- i- Construct the precedence network for the project.
- ii- Determine the critical path and the expected duration to finish all the activities. [47]
- iii- Mention two of the possible reasons for the project to take longer than the expected duration?
- iv- Draw a Gantt chart for the previous project.

Question 3:

For the project shown in the following tables:

Activity	A	B	C	D	E	F	G
Predecessor	----	A	B	A	D	E	----
Duration (weeks)	6	1	3	2	4	10	2

(Cont.)

Activity	H	J	K	L	M	P
Predecessor	G	H , K	----	A	C , L	J, M
Duration (weeks)	10	6	13	8	6	6

- Construct an Activity-On-Arrow (AOA) and an Activity-on-Arrow (AON & AOA).
- Use the critical path method to find the duration of the project. What is the critical path? [ALMP]
- What is the earliest number of weeks in which the project can be finished? [26 weeks]
- If you want to finish the project 2 weeks earlier, would you reduce the duration of “J” or of “E” by 2 weeks? [None]
- What are the allowable delays of “E” and “P”? [4,0]
- Construct a Gantt chart for the project.

Question 4:

Consider the following project:

Activity	Duration (days)	Successor
A	3	B, D
B	8	F
C	4	A, D
D	4	F
E	5	A, D, R
F	4	----
G	7	B, C, E
H	6	G, K
K	1	D, E

Activity	Duration (days)	Successor
L	5	----
M	3	P, W
N	6	L
P	2	S, U
R	6	N
S	3	R, L
T	2	L
U	6	N, R, T
W	5	U

- Use the CPM to find the critical path. [LNREKH]
- Find the total completion time of the project. [35]
- Which activities could be delayed without delaying the early start of the following activities?
- Draw a Gantt chart for the project using a suitable scale. Highlight the critical path and demonstrate the slacks.



Question 5:

Consider the following project:

Activity	Successor	Duration (weeks)
A	D, C	4
B	E	7
C	F	6
D	G	8
E	F, H	6
F	K, X	3
G	J	7
H	K	4
X	L	2
J	L	6
K	M	5
L	---	3
M	---	4

- i- Draw a precedence diagram for the project and use the CPM to find the total completion time. [28]
- ii- Suppose that after 8 weeks, activities A, B, and C have been finished; activity D is 75% finished; and activity E is half finished. How many weeks after the original start time should the project be finished? [26]
- iii- Draw a Gantt chart for the project original schedule.

All final answers are given for guidance only. A student should present a complete answer including the units.

Sheet 4

Program Evaluation and Review Technique (PERT)*

*Normal distribution tables are attached at the end of the sheet

Question 1:

A program of seven projects has probabilistic time estimations as given in the following table.

Activity (i-j)	1-2	1-3	1-4	2-5	3-5	4-6	5-6
Optimistic time (weeks)	2	1	2	1	1	2	3
Most Likely Time (weeks)	5	1	2	4	1	5	6
Pessimistic Time (weeks)	14	1	8	7	7	8	15

- i- Which activities are skewed to the right? Why? [(1-2), (1-4), (3-5), (5-6)]
- ii- How is the skewness of activities (2-5), and (4-6)? [centered]
- iii- What is the expected completion time of the program? What is its standard deviation? [17, 3]

For the following questions, consider only the critical path in calculations

- iv- What is the probability that the program will be finished at least 4 weeks earlier than expected? [9.18]
- v- What is the probability that the program will be completed no later than 18 weeks? [62.93]
- vi- If the program due date is after 19 weeks, what is the probability of not meeting the due date? [25.14]
- vii- On what due date, the probability of meeting the due completion date is 85%? [20.1]
- viii- Repeat questions iv, v, vi considering all paths. [1.72 – 55.93 - 29]

Question 2:

What is the expected completion time of the following project? What is its standard deviation? [14, 4.67]

Activity	A	B	C	D	E	F	G	H
Pred.	—	—	A	B	C	C	D	F
Optimistic time (weeks)	3	2	1	1	1	1	1	1
Most Likely Time (weeks)	4	3	2	4	2	2	2	4
Pessimistic Time (weeks)	11	4	3	7	3	9	3	7

For the following questions, consider all paths in calculations

- i- What is the probability of the project requiring longer than 16 weeks to be completed? [????]
- ii- What is the probability of the project requiring less than 13 weeks to be completed? [40.71]
- iii- What is the probability of finishing within 13 to 16 weeks? [????]

Question 3:

For the following project of probabilistic time estimations,

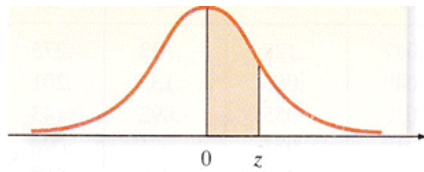
Start node	End node	Activity	Most Likely Time (days)	Optimistic time (days)	Pessimistic Time (days)
1	2	A	10	7	20
1	3	B	5	4.5	7
1	4	C	8	6	10.5
1	5	D	6	4	7.5
2	7	F	6	4	8
3	7	G	8	6.5	11
3	8	H	5	4	7
4	8	J	6	4	7
5	8	K	1	1	2
7	10	O	5	4.5	7
8	10	Z	6	4	9

- i- In how many days is the project expected to be completed? What is the standard deviation of the obtained value? [22.42 , 2.3]

For the following questions, consider only the critical path in calculations

- ii- What is the probability that the project is completed no later than 2 days after the expected? [80.78]
iii- What is the probability that the project will be completed no later than 5 days earlier than expected? [1.5]

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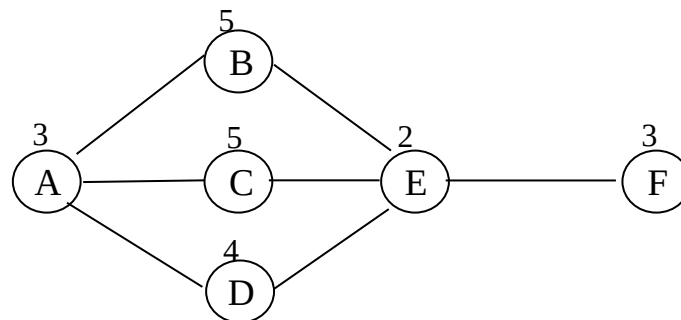
z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
.0	.0000	.0040	.0080	.0120	.0160	.0199	.0239	.0279	.0319	.0359
.1	.0398	.0438	.0478	.0517	.0557	.0596	.0636	.0675	.0714	.0753
.2	.0793	.0832	.0871	.0910	.0948	.0987	.1026	.1064	.1103	.1141
.3	.1179	.1217	.1255	.1293	.1331	.1368	.1406	.1443	.1480	.1517
.4	.1554	.1591	.1628	.1664	.1700	.1736	.1772	.1808	.1844	.1879
.5	.1915	.1950	.1985	.2019	.2054	.2088	.2123	.2157	.2190	.2224
.6	.2257	.2291	.2324	.2357	.2389	.2422	.2454	.2486	.2517	.2549
.7	.2580	.2611	.2642	.2673	.2704	.2734	.2764	.2794	.2823	.2852
.8	.2881	.2910	.2939	.2967	.2995	.3023	.3051	.3078	.3106	.3133
.9	.3159	.3186	.3212	.3238	.3264	.3289	.3315	.3340	.3365	.3389
1.0	.3413	.3438	.3461	.3485	.3508	.3531	.3554	.3577	.3599	.3621
1.1	.3643	.3665	.3686	.3708	.3729	.3749	.3770	.3790	.3810	.3830
1.2	.3849	.3869	.3888	.3907	.3925	.3944	.3962	.3980	.3997	.4015
1.3	.4032	.4049	.4066	.4082	.4099	.4115	.4131	.4147	.4162	.4177
1.4	.4192	.4207	.4222	.4236	.4251	.4265	.4279	.4292	.4306	.4319
1.5	.4332	.4345	.4357	.4370	.4382	.4394	.4406	.4418	.4429	.4441
1.6	.4452	.4463	.4474	.4484	.4495	.4505	.4515	.4525	.4535	.4545
1.7	.4554	.4564	.4573	.4582	.4591	.4599	.4608	.4616	.4625	.4633
1.8	.4641	.4649	.4656	.4664	.4671	.4678	.4686	.4693	.4699	.4706
1.9	.4713	.4719	.4726	.4732	.4738	.4744	.4750	.4756	.4761	.4767
2.0	.4772	.4778	.4783	.4788	.4793	.4798	.4803	.4808	.4812	.4817
2.1	.4821	.4826	.4830	.4834	.4838	.4842	.4846	.4850	.4854	.4857
2.2	.4861	.4864	.4868	.4871	.4875	.4878	.4881	.4884	.4887	.4890
2.3	.4893	.4896	.4898	.4901	.4904	.4906	.4909	.4911	.4913	.4916
2.4	.4918	.4920	.4922	.4925	.4927	.4929	.4931	.4932	.4934	.4936
2.5	.4938	.4940	.4941	.4943	.4945	.4946	.4948	.4949	.4951	.4952
2.6	.4953	.4955	.4956	.4957	.4959	.4960	.4961	.4962	.4963	.4964
2.7	.4965	.4966	.4967	.4968	.4969	.4970	.4971	.4972	.4973	.4974
2.8	.4974	.4975	.4976	.4977	.4977	.4978	.4979	.4979	.4980	.4981
2.9	.4981	.4982	.4982	.4983	.4984	.4984	.4985	.4985	.4986	.4986
3.0	.4987	.4987	.4987	.4988	.4988	.4989	.4989	.4989	.4990	.4990
3.1	.49903	.49906	.49910	.49913	.49916	.49918	.49921	.49924	.49926	.48829
3.2	.49931	.49934	.49936	.49938	.49940	.49942	.49944	.49946	.49948	.49950
3.3	.49952	.49953	.49955	.49957	.49958	.49960	.49961	.49962	.49964	.49965
3.4	.49966	.49968	.49969	.49970	.49971	.49972	.49973	.49974	.49975	.49976
3.5	.49977	.49978	.49978	.49979	.49980	.49981	.49981	.49982	.49983	.49983
3.6	.49984	.49985	.49985	.49986	.49986	.49987	.49987	.49988	.49988	.49989
3.7	.49989	.49990	.49990	.49990	.49991	.49991	.49992	.49992	.49992	.49992
3.8	.49993	.49993	.49993	.49994	.49994	.49994	.49994	.49995	.49995	.49995
3.9	.49995	.49995	.49996	.49996	.49996	.49996	.49996	.49996	.49997	.49997

Sheet 5 Crashing

Question 1:

Before the start of project “XX”, the project planner sent a memo to his boss, the director project management, stating that the XX project would require thirteen weeks for completion according to the figure shown below. The project planner realized that the customer wanted the job completed in less time. After discussion with the functional managers, he/she developed the table shown below:

Act		A	B	C	D	E	F
Normal	Time	3	5	5	4	2	3
	Cost	6,000	12,000	16,000	8,000	6,000	14,000
Crash	Time	2	4	3	2	1	1
	Cost	8,000	13,500	22,000	10,000	7,500	20,000



Find the following:

- The project completion time. [13]
- The most economical crashing plan for this project; time and cost [8 – 76,000]
- The difference in project cost when finished in the normal and the crash time [14,000]
- The minimum cost of finishing the project in nine weeks. [71,500]

Question 2:

Consider the project in the next table. Find the completion time, and the critical path. Then, find the least possible time to complete the project with the least additional cost. Find the project total cost in the latter case. If the manager is ready to pay only EGP 500 to crash the project, what will be the shortest project time?

[31, B-F-G-I-J, 24, 6950, 28]

Question 3:

The data in the table were obtained from a study of the times required to overhaul a chemical plant. Times are in weeks. Costs are in thousand Egyptian pounds.

Find the following:

- The project normal completion time (use AOA), and cost [31-31,000]
- The minimum completion time with the minimum possible cost. [14-47,100]
- The difference in project cost when finished in normal and in crash time. [16,100]



- d) How to reduce the project duration by 10 weeks for the least cost? [F = 6, C = 4, D = 1]
e) The crashing time that minimizes the daily costs if the weekly indirect costs are EGP 750 [9]

Table of Question 2

Activity	Predecessor	Time (days)		Cost (EGP)	
		Normal Time	Crash Time	Normal Cost	Crash Cost
A	---	2	2	100	100
B	---	4	3	400	500
C	A	3	1	1000	1500
D	B	1	1	500	500
E	C	6	5	300	500
F	B	9	7	1500	1900
G	F	7	6	400	700
H	E, D	3	3	50	50
I	G	2	2	150	150
J	H, I	9	6	400	850
K	B	2	2	600	600
L	K	5	5	200	200
M	L	6	6	100	100

Table of Question 3

Activity	Nodes	Crash Schedule		Normal Schedule	
		Time	Cost	Time	Cost
A	1-2	3	6	5	4
B	1-3	1	5	5	3
C	2-4	5	7	10	4
D	3-4	2	6	7	4
E	2-6	2	5	6	3
F	4-6	5	9	11	6
G	4-5	4	6	6	3
H	6-7	1	4	5	2
I	5-7	1	5	4	2

All final answers are given for guidance only. A student should present a complete answer including the units.

Sheet 6

Tools for Managing Project Activities

Question 1:

Construct a process decision program chart (PCPD) for the task of “Preparing the list of attendees invited”.

Question 2:

The following assignments should be submitted during the next 14 days. The table below shows the grades for each assignment as well as the submission deadline.

Assignment	Grade	Deadline
Assignment A	10	12 th October
Assignment B	5	13 th October
Assignment C	7	29 th October
Assignment D	8	20 th October
Assignment E	15	25 th October

Use the six-sigma prioritization matrix to detect the order of doing the assignments. [ADEBC/AEDBC]

Question 3:

In our faculty, some statistics were carried concerning the number of students in different departments, and in different studying years. The study also considered the number of male and female students. It is required to represent the relationships between the following three parameters in one chart.

	Mechanical	Electrical	Architecture	Civil
Sophomore	500	730	220	400
Junior	400	700	200	300
Senior 1	370	690	190	250
Senior 2	350	680	180	220

	Mechanical	Electrical	Architecture	Civil
Male	1000	1700	350	750
Female	620	1100	440	420

	Sophomore	Junior	Senior 1	Senior 2
Male	1100	950	900	850
Female	750	650	600	580



Question 4:

In a project of arranging a sports day in our faculty, make an affinity diagram for the task of “Making invitations for the sports day”. Note that there are two types of invitations: online invitations and paper invitations.

GUIDE: List the performance indicators first, and then categorize them.

Question 5 :

The distances (in Km) between different locations in a construction project were represented in the following table.

	Concrete production	Construction site	Engineers' offices	Buffet	Sales office
Concrete production	-----	1	2	2.5	5
Construction site	1	-----	1	2	4
Engineers' offices	2	1	-----	0.5	2
Buffet	2.5	2	0.5	-----	1
Sales office	5	4	2	1	-----

Use a more convenient matrix to represent the previous relationships.

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